
Nonparametric Regression and Splines

Lecture 4 ■ POLSCI 689L

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Overview

1. **Density estimation:** local counts, kernels, smoothing, and consistency
2. **Regression:** conditional means, Nadaraya–Watson, and local linear regression
3. **Statistical results:** bias, variance, rates, and inference
4. **Series methods:** basis expansions, approximation, estimation, and model selection
5. **Splines:** regression splines, natural splines, and smoothing splines

The first target is an unknown density

Suppose $X_1, \dots, X_n$ are i.i.d. draws from a continuous distribution. Its density $f(x)$ describes where observations are concentrated:

$$\Pr\{a < X \leq b\} = \int_a^b f(v) dv, \quad f(x) \geq 0, \quad \int f(x) dx = 1.$$

If $F(x) = \Pr\{X \leq x\}$ is differentiable, then $f(x) = F'(x)$.

- For wages, the density describes common wage levels, dispersion, and modes.
- A density value is probability per unit of X ; interval probabilities are areas.
- We estimate f without specifying a family such as the normal distribution.

Hansen, *Probability and Statistics for Economists* (2022), Chapter 17; abbreviated P&S below.

The empirical CDF is consistent by the LLN

Fix $x \in \mathbb{R}$. The empirical distribution function is an average of indicators:

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n Z_i(x), \quad Z_i(x) := \mathbf{1}\{X_i \leq x\}.$$

Since the X_i are i.i.d., the $Z_i(x)$ are i.i.d. Bernoulli variables with success probability $F(x)$. Their expectation is

$$\mathbb{E}[Z_i(x)] = 1 \cdot \Pr\{X_i \leq x\} + 0 \cdot \Pr\{X_i > x\} = F(x).$$

Also, $\mathbb{E}|Z_i(x)| = F(x) \leq 1$, so the integrability condition for the LLN holds. Applying the weak law of large numbers gives

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n Z_i(x) \xrightarrow{p} \mathbb{E}[Z_i(x)] = F(x).$$

The SLLN also gives $F_n(x) \xrightarrow{\text{a.s.}} F(x)$ for every fixed x , under the same conditions.

P&S, Theorems 7.4 (WLLN) and 7.12 (SLLN); Tsybakov, §1.2, p. 2. This pointwise result needs neither continuity of F nor finite moments of X_i .

Local counts estimate probability per unit of length

Following Tsybakov, approximate $f(x) = F'(x)$ by a symmetric difference:

$$\frac{F(x+h) - F(x-h)}{2h} = \frac{1}{2h} \int_{x-h}^{x+h} f(v) dv \longrightarrow f(x)$$

when f is continuous at x . Substituting the empirical CDF F_n , whose pointwise consistency follows from the LLN, yields the local-count density estimate

$$\hat{f}_{\text{window}}(x) = \frac{F_n(x+h) - F_n(x-h)}{2h} = \frac{1}{2nh} \sum_i \mathbf{1}\{x-h < X_i \leq x+h\}.$$

The estimate jumps when an observation enters or leaves the window. A continuous weighting function will smooth these changes.

Consistency of this density estimate requires a separate bandwidth argument: the evaluation points $x \pm h_n$ move with n , and the denominator $2h_n$ shrinks.

Tsybakov, *Introduction to Nonparametric Estimation*, §1.2, pp. 2–3. We write f for the density denoted p in Tsybakov.

Kernels and their order

Replace the abrupt window by weights $K((X_i - x)/h)$. Following Tsybakov, a **kernel** is a function $K : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$\int |K(u)| du < \infty, \quad \int K(u) du = 1.$$

For an integer $\ell \geq 1$, it is a **kernel of order ℓ** if, in addition,

$$\int |u|^j |K(u)| du < \infty, \quad \mu_j(K) := \int u^j K(u) du = 0, \quad j = 1, \dots, \ell.$$

- Note that different people may define the order slightly different. (We will see this later)
- Total mass is one, while the first ℓ moments vanish.
- This definition allows signed and asymmetric kernels.
- The $(\ell + 1)$ st moment need not be nonzero, or even exist. A kernel of order ℓ therefore also has every lower positive order.

Tsybakov, §1.2, p. 3; Definition 1.3, p. 5, and the discussion on p. 6. Kernel order refers to moment conditions.

The kernel density estimator averages local weights

For our nonnegative kernels K and bandwidth $h > 0$, the kernel density estimator (KDE) is

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{X_i - x}{h}\right).$$

Each observation contributes a density centered at X_i , with width set by h . The factor $1/h$ preserves its area when we change its width. Since $K \geq 0$, we have $\hat{f}(x) \geq 0$; also,

$$\int \hat{f}(x) dx = \frac{1}{n} \sum_{i=1}^n \int \frac{1}{h} K\left(\frac{X_i - x}{h}\right) dx = \frac{1}{n} \sum_{i=1}^n \int K(u) du = 1.$$

- A small bandwidth produces narrow local features and a more variable estimate.
- A large bandwidth spreads each observation's contribution over a wider region.

Tsybakov, §1.2, equation (1.2); P&S, §17.3, equation (17.2), and the normalization argument on p. 340.

Density consistency: the bias vanishes

Fix x . Let $X_1, \dots, X_n$ be i.i.d. with density f , continuous at x and bounded by $M < \infty$. Assume

$$\int K(u) du = 1, \quad \int |K(u)| du < \infty, \quad R_K := \int K(u)^2 du < \infty.$$

Choose $h = h_n \rightarrow 0$ and $nh_n \rightarrow \infty$. Set $U_i = (X_i - x)/h$; Since the X_i are identically distributed,

$$\begin{aligned} \mathbb{E}[\hat{f}(x)] &= \frac{1}{nh} \sum_{i=1}^n \mathbb{E}\left[K\left(\frac{X_i - x}{h}\right)\right] = \frac{1}{h} \mathbb{E}\left[K\left(\frac{X_1 - x}{h}\right)\right] = \int \frac{1}{h} K\left(\frac{X_1 - x}{h}\right) f(x) dx \\ &= \int K(u) f(x + hu) du = f(x) + \int K(u)(f(x + hu) - f(x)) du. \end{aligned}$$

For each u , $f(x + hu) \rightarrow f(x)$;

$$\mathbb{E}[\hat{f}(x)] = f(x) + o(1)$$

P&S, Theorem 17.6, using sufficient bounded-density conditions that also cover Gaussian kernels. For bounded, compact-support kernels, continuity near x suffices without a global bound on f .

Density consistency and the NW denominator

Independence and $\text{Var}(Z) \leq \mathbb{E}[Z^2]$ give

$$\begin{aligned}\text{Var}(\hat{f}(x)) &= \frac{1}{nh^2} \text{Var}\left(K\left(\frac{X_1 - x}{h}\right)\right) \\ &\leq \frac{1}{nh} \int K(u)^2 f(x + hu) du \leq \frac{MR_K}{nh} \rightarrow 0.\end{aligned}$$

Combining this with $b_n(x) \rightarrow 0$, for every $\varepsilon > 0$,

$$\Pr\{|\hat{f}(x) - f(x)| > \varepsilon\} \leq \frac{\mathbb{E}[(\hat{f}(x) - f(x))^2]}{\varepsilon^2} = \frac{\text{Var}(\hat{f}(x)) + b_n(x)^2}{\varepsilon^2} \rightarrow 0.$$

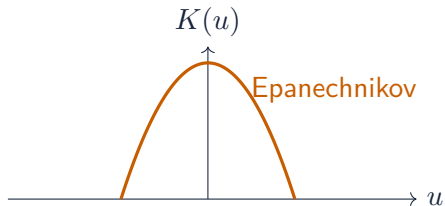
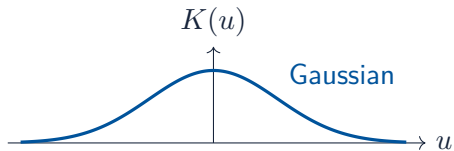
Thus $\hat{f}(x) \xrightarrow{p} f(x)$. If $f(x) > 0$, then $\Pr\{\hat{f}(x) > f(x)/2\} \rightarrow 1$ and the continuous mapping theorem yields

$$\boxed{\hat{f}(x)^{-1} = f(x)^{-1} + o_p(1).}$$

This is the denominator convergence used in the NW consistency argument.

Tsybakov, Proposition 1.1, for the variance bound; P&S, Theorem 17.6. This is pointwise consistency at a fixed x .

Gaussian and Epanechnikov kernels



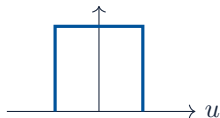
Common kernels have simple functional forms

Kernel	Function $K(u)$	Support
Uniform	$\frac{1}{2}\mathbf{1}\{ u \leq 1\}$	$[-1, 1]$
Triangular	$(1 - u)\mathbf{1}\{ u \leq 1\}$	$[-1, 1]$
Epanechnikov	$\frac{3}{4}(1 - u^2)\mathbf{1}\{ u \leq 1\}$	$[-1, 1]$
Biweight (quartic)	$\frac{15}{16}(1 - u^2)^2\mathbf{1}\{ u \leq 1\}$	$[-1, 1]$
Gaussian	$\frac{1}{\sqrt{2\pi}} \exp(-u^2/2)$	$\mathbb{R}$

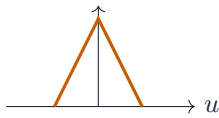
Normalization

These standard forms integrate to one. Their second moments need not equal one; the regression results later use rescaled kernels with $\mu_2(K) = 1$.

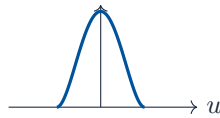
Kernel shapes differ in support and smoothness



Uniform
discontinuous cutoff



Triangular
linear decay, kink at zero

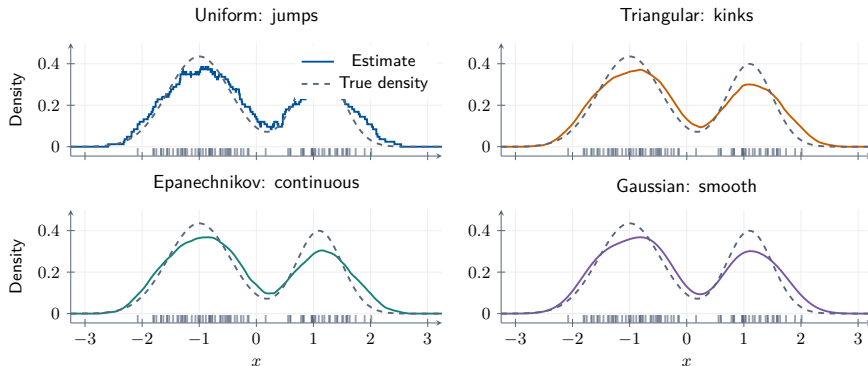


Biweight
smooth compact support

- Compact-support kernels assign exactly zero weight outside $|X_i - x| \leq h$; the Gaussian kernel assigns a positive, rapidly vanishing weight everywhere.
- Once h is calibrated, these kernels usually produce similar density estimates; bandwidth choice matters much more.

Kernel shape and density smoothness

Same sample ($n = 80$), same $h = 0.30$; all kernels are rescaled to $\mu_2(K) = 1$.

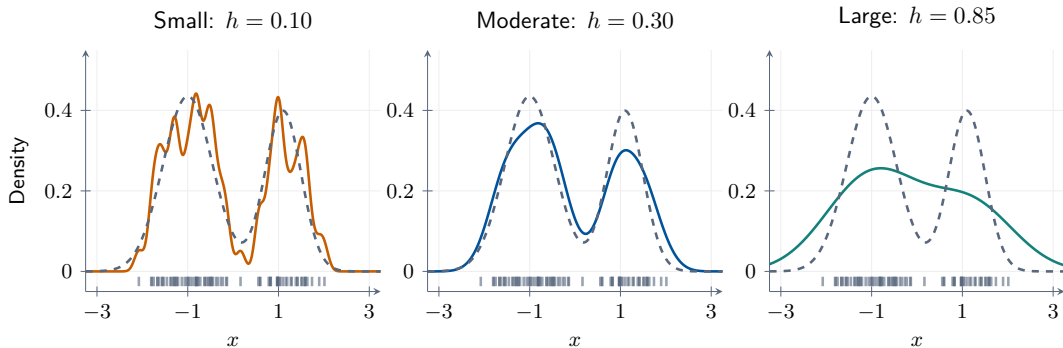


The kernel determines jumps and kinks; the broad features are similar here.

Original simulation: $0.6N(-1, 0.55^2) + 0.4N(1.1, 0.4^2)$, seed 20260914. For a standard kernel with second moment s^2 , use $sK(su)$; each contribution then has standard deviation h .

Bandwidth and density smoothing

Same observations and Gaussian kernel throughout. Dashed curves show the true density.



A small h reveals sample noise; a large h blurs the two modes together.

Same simulated sample as the preceding slide ($n = 80$); ticks along the horizontal axis mark observations. Bandwidths are illustrative, not selected by cross-validation.

Kernel estimation of a joint density

Now observe i.i.d. pairs (X_i, Y_i) with joint density $p(x, y)$. From the univariate kernel K , construct the **product kernel**

$$\mathcal{K}(u, v) = K(u)K(v), \quad \iint \mathcal{K}(u, v) du dv = 1.$$

Using the same bandwidth $h > 0$ in both coordinates, define

$$\hat{p}(x, y) = \frac{1}{nh^2} \sum_{i=1}^n K\left(\frac{X_i - x}{h}\right) K\left(\frac{Y_i - y}{h}\right).$$

- Each observed pair contributes a two-dimensional density centered at (X_i, Y_i) .
- The factor $1/h^2$ accounts for rescaling both coordinates.
- A product *kernel* does not assume that X and Y are independent: each summand uses the two coordinates of the same observed pair.

Tsybakov, §1.2, equation (1.3), pp. 3–4. We retain the nonnegative kernels introduced above.

The joint KDE recovers the marginal KDE

The population marginal density is $f(x) = \int p(x, y) dy$. The joint estimator has the same property. With $u = (Y_i - y)/h$,

$$\int K\left(\frac{Y_i - y}{h}\right) dy = h \int K(u) du = h.$$

Integrating the product estimator over y therefore gives

$$\begin{aligned} \int \hat{p}(x, y) dy &= \frac{1}{nh^2} \sum_i K\left(\frac{X_i - x}{h}\right) \int K\left(\frac{Y_i - y}{h}\right) dy \\ &= \frac{1}{nh} \sum_i K\left(\frac{X_i - x}{h}\right) = \hat{f}(x). \end{aligned}$$

Thus the joint and marginal estimates are compatible, and $\iint \hat{p}(x, y) dx dy = \int \hat{f}(x) dx = 1$.

Direct integration of Tsybakov's equations (1.2)–(1.3). The marginal KDE uses the same X kernel and bandwidth as the joint KDE.

Many substantive relationships need not be linear

Economic and political behavior

- How do wages evolve with labor-market experience?
- Does public spending have diminishing returns?
- Does political participation respond differently at low and high income levels?

Institutions and interventions

- Does class size matter differently for initially weak and strong students?
- Are there thresholds in the relationship between debt and growth?
- Where does a dose-response curve flatten or change direction?

Research question

Can we let the data reveal the shape of a relationship while still controlling overfitting and quantifying uncertainty?

The target is an unknown conditional mean

Suppose (Y_i, X_i) are independent draws and X is initially scalar. Write

$$Y = m(X) + e, \quad m(x) = \mathbb{E}[Y \mid X = x], \quad \mathbb{E}[e \mid X] = 0.$$

The conditional variance is

$$\sigma^2(x) = \mathbb{E}[e^2 \mid X = x] = \text{Var}(Y \mid X = x).$$

- Prediction asks for a good estimate $\hat{m}(x)$ at new values of x .
- Description asks how the conditional mean changes over the support of X .
- Inference asks how much sampling uncertainty and smoothing bias remain.

Key idea

The object is a function, not a finite list of coefficients.

Nonparametric does not mean assumption-free

Parametric regression

Choose a fixed functional form such as

$$m(x) = \beta_0 + \beta_1 x + \beta_2 x^2.$$

The number of parameters stays fixed as n grows.

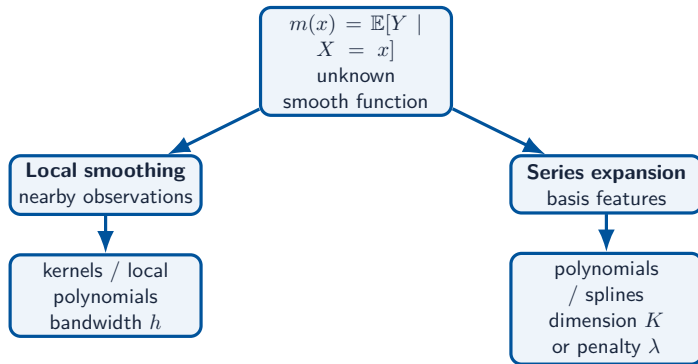
Nonparametric regression

Allow complexity to grow with the sample.
Replace a rigid functional form with smoothness, locality, or approximation assumptions.

Key idea

We exchange a strong shape restriction for a tuning problem: how flexible should the estimated function be?

Two strategies learn the same unknown function



Key idea

Both strategies reproduce the familiar bias–variance trade-off in different notation.

Binned means provide the starting intuition

For discrete X , estimate $m(x)$ by averaging outcomes among observations with $X_i = x$.
With a continuously distributed X , there may be no observations satisfying $X_i = x$.
Average observations in a neighborhood instead:

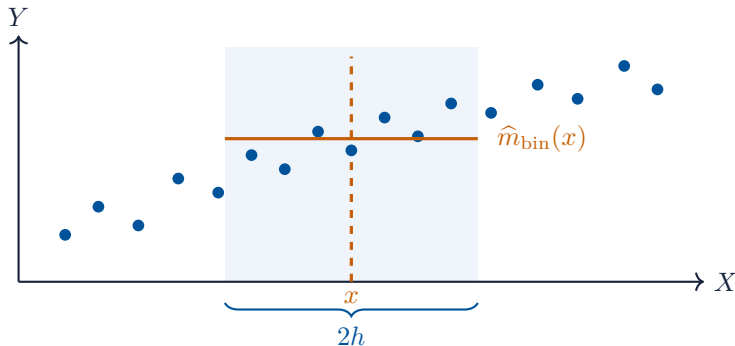
$$\hat{m}_{\text{bin}}(x) = \frac{\sum_{i=1}^n \mathbf{1}\{|X_i - x| \leq h\} Y_i}{\sum_{i=1}^n \mathbf{1}\{|X_i - x| \leq h\}}.$$

- The bandwidth h determines which observations count as nearby.
- A rolling window avoids committing to one arbitrary partition of the support.
- Equal weights create abrupt changes when observations enter or leave the window.

Key idea

Density estimation counts nearby observations; regression averages their outcomes.

A bandwidth defines what counts as local



Small h

Few observations: more local detail, but a noisy estimate.

Large h

Many observations: lower variance, but local features are averaged away.

Nadaraya–Watson is a weighted local mean

The density plug-in argument gives the Nadaraya–Watson estimator

$$\hat{m}_{\text{NW}}(x) = \frac{\sum_{i=1}^n K\left(\frac{X_i - x}{h}\right) Y_i}{\sum_{i=1}^n K\left(\frac{X_i - x}{h}\right)} = \sum_{i=1}^n w_i(x) Y_i,$$

where

$$w_i(x) = \frac{K((X_i - x)/h)}{\sum_{j=1}^n K((X_j - x)/h)}, \quad \sum_{i=1}^n w_i(x) = 1.$$

Nearby observations receive more weight; distant observations receive little or no weight.

Key idea

The rolling binned means estimator is the NW estimator with the rectangular kernel.

Bandwidth controls the estimator's flexibility

$h \rightarrow 0$

At a training point, one observation dominates:

$$\hat{m}(X_i) \rightarrow Y_i.$$

Training error approaches zero.

Intermediate h

The estimate averages enough observations to suppress noise while preserving systematic curvature.

$h \rightarrow \infty$

All observations receive nearly equal weight:

$$\hat{m}_{\text{NW}}(x) \rightarrow \bar{Y}.$$

Local structure disappears.

Key idea

Bandwidth is an inverse complexity parameter: smaller h means a more flexible fit.

Kernel choice is usually a second-order decision

What changes with the kernel?

- Compact versus unbounded support
- Smoothness of the estimated curve and its derivatives
- An efficiency constant through the kernel roughness $R_K = \int K(u)^2 du$

What changes with the bandwidth?

- Effective local sample size
- Bias and variance
- Whether important features are visible or smoothed away

Practical priority

Choose h carefully. Epanechnikov minimizes the usual asymptotic MISE criterion among nonnegative second-order kernels; standard kernels differ little in this criterion.

Regression as a conditional-density mean

An intuitive motivation of NW estimator:

$$m(x) = \int y p(y | x) dy = \frac{\int y p(x, y) dy}{f(x)}$$

A natural estimate of the regression function is its mean:

$$\tilde{m}(x) := \int y \hat{p}(y | x) dy = \frac{\int y \hat{p}(x, y) dy}{\hat{f}(x)}.$$

We can show that NW estimator is equivalent to this kernel estimator.

Proposition 1.10: integrate the outcome kernel

Tsybakov's **kernel of order 1** satisfies

$$\int K(u) du = 1, \quad \int uK(u) du = 0, \quad \int |u| |K(u)| du < \infty.$$

Our symmetric kernels satisfy these conditions. The key step is to compute

$$J_i := \frac{1}{h} \int y K\left(\frac{Y_i - y}{h}\right) dy.$$

Set $u = (Y_i - y)/h$, so $y = Y_i - hu$ and $dy = -h du$. Reversing the integration limits gives

$$\begin{aligned} J_i &= \int (Y_i - hu) K(u) du \\ &= Y_i \int K(u) du - h \int u K(u) du = Y_i. \end{aligned}$$

Each outcome kernel has mean Y_i : smoothing in y preserves its first moment.

Tsybakov, Proposition 1.10 and proof, pp. 33–34. Hansen calls the symmetric kernels used here second-order kernels.

Proposition 1.10: recovering NW from the joint KDE

Substitute $J_i = Y_i$ into the first moment of the joint KDE:

$$\begin{aligned}\int y \hat{p}(x, y) dy &= \frac{1}{nh} \sum_i K\left(\frac{X_i - x}{h}\right) \left\{ \frac{1}{h} \int y K\left(\frac{Y_i - y}{h}\right) dy \right\} \\ &= \frac{1}{nh} \sum_i K\left(\frac{X_i - x}{h}\right) Y_i.\end{aligned}$$

Using $\hat{f}(x) = (nh)^{-1} \sum_i K((X_i - x)/h)$, the estimated conditional mean becomes

$$\tilde{m}(x) = \frac{\int y \hat{p}(x, y) dy}{\hat{f}(x)} = \frac{\sum_i K((X_i - x)/h) Y_i}{\sum_i K((X_i - x)/h)} = \hat{m}_{\text{NW}}(x).$$

NW therefore arises by estimating the joint density and taking the mean of the implied conditional density. The outcome kernels disappear after integration.

Tsybakov, Proposition 1.10, equation (1.61), and its proof, pp. 33–34. The identity holds for every $h > 0$ whenever $\hat{f}(x) > 0$.

Kernel regression is local weighted least squares

The Nadaraya–Watson estimate also solves

$$\hat{m}_{\text{NW}}(x) = \arg \min_{a \in \mathbb{R}} \sum_{i=1}^n K\left(\frac{X_i - x}{h}\right) (Y_i - a)^2.$$

- The fitted model is locally constant: $m(X_i) \approx a$ near x .
- The target point changes, so the weights and the fitted constant change.
- This optimization view immediately suggests fitting a local line or polynomial instead.

Key idea

Kernel regression is not separate from regression: it is regression with target-specific weights and a deliberately simple local model.

Local linear regression fits a line around each target

Use the first-order approximation

$$m(X_i) \approx m(x) + m'(x)(X_i - x).$$

Estimate its intercept and slope by

$$\begin{pmatrix} \hat{m}_{LL}(x) \\ \hat{m}'_{LL}(x) \end{pmatrix} = \arg \min_{a,b} \sum_{i=1}^n K\left(\frac{X_i - x}{h}\right) [Y_i - a - b(X_i - x)]^2.$$

- The intercept estimates the level $m(x)$.
- The slope estimates the local marginal relationship $m'(x)$.
- As $h \rightarrow \infty$, the estimator approaches global OLS.

The local linear estimate is an ordinary WLS solution

Let

$$Z_i(x) = \begin{pmatrix} 1 \\ X_i - x \end{pmatrix}, \quad W_x = \text{diag} \left\{ K \left(\frac{X_i - x}{h} \right) \right\}_{i=1}^n.$$

Stack $Z_i(x)'$ into the matrix Z_x . Then

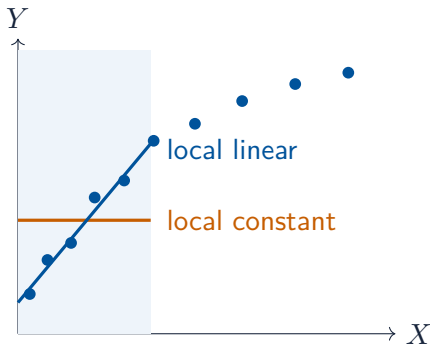
$$\hat{\beta}_{\text{LL}}(x) = (Z_x' W_x Z_x)^{-1} Z_x' W_x \mathbf{Y}, \quad \hat{m}_{\text{LL}}(x) = \begin{pmatrix} 1 & 0 \end{pmatrix} \hat{\beta}_{\text{LL}}(x).$$

Hence $\hat{m}_{\text{LL}}(x) = \sum_i w_i^{\text{LL}}(x) Y_i$.

Key idea

All the familiar weighted least-squares machinery remains available locally.

Local linear regression reduces boundary bias



Near the lower boundary, observations exist only to the right of x .

- Local constant regression treats this asymmetric neighborhood as if the function were flat.
- A local line uses the slope to extrapolate back to the target point.
- Interior bias is $O(h^2)$ for both methods, but NW boundary bias is $O(h)$; local linear boundary bias remains $O(h^2)$.

Default recommendation

Use local linear rather than local constant regression, especially near the support boundary.

Local polynomial regression extends the same idea

Approximate the regression function near x using a degree- p Taylor expansion:

$$m(X_i) \approx \sum_{j=0}^p \frac{m^{(j)}(x)}{j!} (X_i - x)^j.$$

Estimate the coefficients with kernel-weighted least squares.

$p = 0$

Local constant /
Nadaraya–Watson

$p = 1$

Local linear regression

$p \geq 2$

Local curvature and higher
derivatives

Key idea

Increasing polynomial order can reduce approximation bias, but it introduces more locally estimated coefficients and greater variance.

Consistency needs $h \rightarrow 0$ and $nh \rightarrow \infty$

For a fixed interior point x , maintain:

$$h \rightarrow 0, \quad nh \rightarrow \infty, \quad f(x) > 0.$$

In a neighborhood of x , assume $m(\cdot)$, $f(\cdot)$, and $\sigma^2(\cdot)$ are continuous.

$h \rightarrow 0$

The local approximation is evaluated over an increasingly narrow neighborhood, reducing smoothing bias.

$nh \rightarrow \infty$

The expected number of observations in that neighborhood still diverges, reducing sampling noise.

Key idea

Nonparametric consistency is possible because the neighborhood shrinks more slowly than the sample grows.

Local linear removes the design-density bias

Under the bias assumptions and the normalization $\mu_2 = 1$, as $nh \rightarrow \infty$ with $h \rightarrow 0$,

$$\mathbb{E}[\hat{m}_{LL}(x) \mid \mathbf{X}] = m(x) + h^2 B_{LL}(x) + o_p(h^2) + O_p\left(\sqrt{\frac{h}{n}}\right),$$

where

$$B_{LL}(x) = \frac{1}{2}m''(x).$$

- The leading bias $h^2 B_{LL}(x)$ depends on the curvature of the conditional mean, without a design-density term.
- If $m''(x) < 0$, the leading bias is negative.

Hansen, *Econometrics*, Theorem 19.1(2), p. 677. For a kernel without unit second moment, $B_{LL} = \mu_2 m''/2$. Hansen refers to Fan and Gijbels (1996) for the LL proof.

NW bias depends on curvature and design density

Under the same bias assumptions and the normalization $\mu_2 = 1$,

$$\mathbb{E}[\hat{m}_{\text{NW}}(x) \mid \mathbf{X}] = m(x) + h^2 B_{\text{NW}}(x) + o_p(h^2) + O_p\left(\sqrt{\frac{h}{n}}\right),$$

where

$$B_{\text{NW}}(x) = \frac{1}{2}m''(x) + \frac{f'(x)}{f(x)}m'(x).$$

- NW has an additional bias term involving $m'(x)$ and $f'(x)$ because it is a local average.
- If the design density slopes upward, more observations lie on one side of x than the other.
- A nonzero regression slope then converts this design imbalance into additional bias.

Hansen, *Econometrics*, Theorem 19.1(1), p. 677; proof in §19.26, pp. 700–701. General kernels multiply both bias terms by μ_2 .

Assumptions for the pointwise proofs

Let (Y_i, X_i) be i.i.d., $Y_i = m(X_i) + e_i$, and $\mathbb{E}[e_i | X_i] = 0$. Fix an **interior point** x and a deterministic bandwidth sequence $h = h_n$.

- **Assumption 19.1:** $h \rightarrow 0$, $nh \rightarrow \infty$, $f(x) > 0$, and m , f , and σ^2 are continuous in a neighborhood $\mathcal{N}$ of x .
- **For bias:** m'' and f' are continuous in $\mathcal{N}$.
- **Kernel:** $K \geq 0$ is bounded and symmetric, with $\int K(u) du = 1$ and $\mu_2 = \int u^2 K(u) du = 1$.
- **Proof simplification:** $K(u) = 0$ for $|u| > a$ for some finite a . For small h , every observation receiving weight lies in $\mathcal{N}$.

Compact support follows Hansen's technical proofs; extending these arguments to the Gaussian kernel requires additional control of the tails.

Hansen, *Econometrics*, Assumption 19.1 and §19.26. We use the kernel density consistency result established in the opening section.

Probability tools used in the proofs

- **Taylor expansion and dominated convergence:** expand smooth functions at x ; pass limits through integrals when the integrands have an integrable bound.
- **Chebyshev's inequality:** for $t > 0$,

$$\Pr\{|Z_n - \mathbb{E}Z_n| > t\} \leq \frac{\text{Var}(Z_n)}{t^2}.$$

In particular, $\text{Var}(Z_n) = O(a_n^2)$ implies $Z_n - \mathbb{E}Z_n = O_p(a_n)$.

- **Continuous mapping and Slutsky:** if $A_n \rightarrow_p a$ and $D_n \rightarrow_p d \neq 0$, then $A_n/D_n \rightarrow_p a/d$. If $Z_n \rightarrow_d Z$, then $Z_n/D_n \rightarrow_d Z/d$.
- **Lindeberg CLT:** independent triangular-array summands are asymptotically normal when their total variance converges and their large-summand contribution vanishes. We verify this condition below.

P&S: Theorems A.15–A.16 (Taylor), A.26 (dominated convergence), 7.1 (Chebyshev), 7.7 (continuous mapping), 8.7 (Slutsky), and 9.1 (Lindeberg CLT).

NW bias proof: the conditional expectation

Conditional on $\mathbf{X}$, the kernel weights are fixed and $\mathbb{E}[Y_i | \mathbf{X}] = m(X_i)$. Therefore,

$$\mathbb{E}[\hat{m}_{\text{NW}}(x) | \mathbf{X}] - m(x) = \frac{\sum_i K_i \{m(X_i) - m(x)\}}{\sum_i K_i} = \frac{\hat{b}}{\hat{f}(x)},$$

where

$$\hat{b} = \frac{1}{nh} \sum_i K_i \{m(X_i) - m(x)\}.$$

Its expectation is an integral over the regressor density:

$$\begin{aligned} \mathbb{E}\hat{b} &= \frac{1}{h} \int K\left(\frac{v-x}{h}\right) \{m(v) - m(x)\} f(v) dv \\ &= \int K(u) \{m(x+hu) - m(x)\} f(x+hu) du, \end{aligned}$$

where $v = x + hu$ and $dv = h du$ in the second line.

Hansen, *Econometrics*, equations (19.27)–(19.29), p. 700.

NW bias proof: Taylor expansion

Continuity of m'' and f' gives Taylor expansions, uniform on $|u| \leq a$:

$$\begin{aligned}m(x + hu) - m(x) &= hu m'(x) + \frac{h^2 u^2}{2} m''(x) + o(h^2), \\f(x + hu) &= f(x) + hu f'(x) + o(h).\end{aligned}$$

Multiply these expressions and integrate against $K(u)$:

$$\begin{aligned}\widehat{\mathbb{E}b} &= h m'(x) f(x) \int u K(u) du \\&\quad + h^2 \left\{ \frac{1}{2} m''(x) f(x) + m'(x) f'(x) \right\} \int u^2 K(u) du + o(h^2) \\&= h^2 f(x) B_{\text{NW}}(x) + o(h^2).\end{aligned}$$

Here $\int u K(u) du = 0$ by symmetry and $\int u^2 K(u) du = 1$. The product $m'(x) f'(x)$ generates the additional design-density bias.

Hansen, *Econometrics*, equations (19.30)–(19.32); P&S, Taylor's theorem. Only one derivative of f is needed.

NW bias proof: variation across designs

Independence and $\text{Var}(Z) \leq \mathbb{E}[Z^2]$ imply

$$\begin{aligned}\text{Var}(\hat{b}) &= \frac{1}{nh^2} \text{Var}\left(K\left(\frac{X-x}{h}\right) \{m(X) - m(x)\}\right) \\ &\leq \frac{1}{nh} \int K(u)^2 \{m(x+hu) - m(x)\}^2 f(x+hu) du \\ &\leq C \frac{h}{n} \int u^2 K(u)^2 du = O\left(\frac{h}{n}\right).\end{aligned}$$

The last bound uses local boundedness of m' and f , so that $|m(x+hu) - m(x)| \leq C_1 h|u|$ and $f(x+hu) \leq C_2$ on the kernel support. Chebyshev gives, for any $M > 0$,

$$\Pr\left\{|\hat{b} - \mathbb{E}\hat{b}| > M\sqrt{h/n}\right\} \leq \frac{C_3}{M^2}, \quad \hat{b} - \mathbb{E}\hat{b} = O_p\left(\sqrt{h/n}\right).$$

Hansen, *Econometrics*, equation (19.33); P&S, Theorem 7.1 (Chebyshev). The randomness here comes from $\mathbf{X}$, not the errors e_i .

NW bias proof: divide by the estimated density

The previous two steps establish

$$\hat{b} = h^2 f(x) B_{\text{NW}}(x) + o(h^2) + O_p\left(\sqrt{h/n}\right).$$

Since $\hat{f}(x) \rightarrow_p f(x) > 0$, continuous mapping gives $\hat{f}(x)^{-1} = f(x)^{-1} + o_p(1) = O_p(1)$. Consequently,

$$\begin{aligned} \frac{\hat{b}}{\hat{f}(x)} &= h^2 B_{\text{NW}}(x) \frac{f(x)}{\hat{f}(x)} + \frac{o(h^2) + O_p(\sqrt{h/n})}{\hat{f}(x)} \\ &= h^2 B_{\text{NW}}(x) + o_p(h^2) + O_p\left(\sqrt{h/n}\right). \end{aligned}$$

Substituting into the exact identity proves Theorem 19.1(1):

$$\mathbb{E}[\hat{m}_{\text{NW}}(x) \mid \mathbf{X}] = m(x) + h^2 B_{\text{NW}}(x) + o_p(h^2) + O_p\left(\sqrt{h/n}\right).$$

Hansen, *Econometrics*, equations (19.27), (19.34); P&S, Theorems 17.6 and 7.7. This is a conditional-bias expansion, not an expectation of a ratio replaced by a ratio of expectations.

NW and local linear share the leading variance

Under Assumption 19.1, define the kernel roughness

$$R_K = \int K(u)^2 du.$$

For Nadaraya–Watson and local linear regression,

$$\text{Var}(\hat{m}(x) \mid \mathbf{X}) = \frac{R_K \sigma^2(x)}{nhf(x)} + o_p\left(\frac{1}{nh}\right).$$

- nh is the effective local sample size.
- Larger conditional noise $\sigma^2(x) = \mathbb{E}[e^2 \mid X = x]$ means greater uncertainty.
- Smaller predictor density $f(x)$ means fewer useful observations and greater uncertainty.
- A rougher kernel has larger R_K and slightly larger variance.

Hansen, *Econometrics*, Theorem 19.2, p. 679. These are interior-point, conditional variances.

NW variance proof: conditional variance

Conditional on $\mathbf{X}$, the errors are independent, with mean zero and variance $\sigma^2(X_i)$. The NW weights are $w_i = K_i / \{nh\hat{f}(x)\}$, so

$$\begin{aligned}\text{Var}(\hat{m}_{\text{NW}}(x) \mid \mathbf{X}) &= \sum_i w_i^2 \sigma^2(X_i) = \frac{\hat{v}}{nh\hat{f}(x)^2}, \\ \hat{v} &:= \frac{1}{nh} \sum_i K_i^2 \sigma^2(X_i).\end{aligned}$$

To find the numerator's expectation, change variables:

$$\mathbb{E}\hat{v} = \int K(u)^2 \sigma^2(x + hu) f(x + hu) du \longrightarrow R_K \sigma^2(x) f(x).$$

Continuity and compact support justify dominated convergence: locally bounded $\sigma^2 f$ times K^2 supplies an integrable bound.

Hansen, *Econometrics*, equation (19.4) and proof of Theorem 19.2(1); P&S, Theorem A.26 (dominated convergence).

NW variance proof: convergence of the numerator

Independence and a second-moment bound give

$$\begin{aligned}\text{Var}(\hat{v}) &\leq \frac{1}{nh} \int K(u)^4 [\sigma^2(x + hu)]^2 f(x + hu) du \\ &= O\left(\frac{1}{nh}\right) = o(1).\end{aligned}$$

Bounded K and locally bounded σ^2, f make the integrals finite. Chebyshev and the expectation limits yield

$$\hat{v} \xrightarrow{p} R_K \sigma^2(x) f(x).$$

Since $\hat{f}(x) \rightarrow_p f(x) > 0$, continuous mapping completes the NW proof:

$$nh \text{Var}(\hat{m}_{\text{NW}}(x) \mid \mathbf{X}) = \frac{\hat{v}}{\hat{f}(x)^2} \xrightarrow{p} \frac{R_K \sigma^2(x)}{f(x)}.$$

Hansen, *Econometrics*, Theorem 19.2(1); P&S, Theorems 7.1 and 7.7. The square $[\sigma^2(\cdot)]^2$ is not $\mathbb{E}[e^4 \mid X = \cdot]$; no fourth error moment is assumed.

Kernel estimators are normal around a biased center

Maintain the bias assumptions. Suppose additionally that, for some $r > 2$, $\sup_{v \in \mathcal{N}} \mathbb{E}[|e|^r \mid X = v] < \infty$ and $nh^5 = O(1)$. Then, for NW and LL,

$$\sqrt{nh} [\hat{m}(x) - m(x) - h^2 B(x)] \xrightarrow{d} N\left(0, \frac{R_K \sigma^2(x)}{f(x)}\right).$$

- The convergence rate is $\sqrt{nh}$, not $\sqrt{n}$.
- The first-order distribution contains smoothing bias.

Hansen, *Econometrics*, Theorem 19.9, p. 688. Here $B = B_{\text{NW}}$ or B_{LL} as appropriate. The normal limit is nondegenerate if $\sigma^2(x) > 0$.

Undersmoothing makes bias asymptotically negligible

The centered statistic is bias-free at first order if

$$\sqrt{nh} h^2 \rightarrow 0 \iff nh^5 = o(1).$$

Equivalently, choose $h = o(n^{-1/5})$. Then

$$\sqrt{nh} [\hat{m}(x) - m(x)] \xrightarrow{d} N\left(0, \frac{R_K \sigma^2(x)}{f(x)}\right).$$

- The condition does not literally remove finite-sample bias.
- It does not supply an automatic empirical rule for choosing an inference bandwidth.

Key idea

Undersmoothing is a theoretical device with a real efficiency cost, not a free correction.

Pointwise mean squared error displays the trade-off

Squared leading bias plus leading variance gives

$$\text{AMSE}\{\hat{m}(x)\} = \underbrace{h^4 B(x)^2}_{\text{squared bias}} + \underbrace{\frac{R_K \sigma^2(x)}{nh f(x)}}_{\text{variance}}.$$

Decrease h

Bias falls at rate h^2 , but variance rises because fewer observations receive substantial weight.

Increase h

Variance falls, but the local approximation averages over more curvature.

Key idea

The same bandwidth cannot minimize bias and variance separately; it must balance them.

Integrated risk gives the bandwidth rate

Let $w(x)$ be a trimming or importance weight and define

$$\mathcal{B} = \int B(x)^2 f(x) w(x) dx, \quad \mathcal{V} = \int \sigma^2(x) w(x) dx.$$

Then

$$\text{AIMSE}(h) = h^4 \mathcal{B} + \frac{R_K \mathcal{V}}{nh}.$$

Differentiating with respect to h yields

$$h_{\text{opt}} = \left(\frac{R_K \mathcal{V}}{4\mathcal{B}} \right)^{1/5} n^{-1/5}, \quad \text{AIMSE}(h_{\text{opt}}) = O(n^{-4/5}) \approx 1.65(R_K^4 \mathcal{B} \mathcal{V}^4)^{1/5} n^{-4/5}.$$

So, the optimal kernel here is Epanechnikov kernel that has the smallest R_K .

Deriving the optimal bandwidth

Starting from

$$\text{AIMSE}(h) = h^4 \mathcal{B} + \frac{R_K \mathcal{V}}{nh},$$

differentiate with respect to h :

$$4h^3 \mathcal{B} - \frac{R_K \mathcal{V}}{nh^2} = 0.$$

Hence

$$h^5 = \frac{R_K \mathcal{V}}{4n\mathcal{B}}, \quad h_{\text{opt}} = \left(\frac{R_K \mathcal{V}}{4\mathcal{B}} \right)^{1/5} n^{-1/5}.$$

At this value,

$$\underbrace{\frac{R_K \mathcal{V}}{nh_{\text{opt}}}}_{\text{integrated variance}} = 4 \underbrace{h_{\text{opt}}^4 \mathcal{B}}_{\text{integrated squared bias}}.$$

At the MSE-optimal bandwidth, bias is first-order

If $h \asymp n^{-1/5}$, then

$$h^2 \asymp n^{-2/5}, \quad (nh)^{-1/2} \asymp n^{-2/5}.$$

Therefore

$$\underbrace{h^2 B(x)}_{\text{smoothing bias}} \quad \text{and} \quad \underbrace{\text{se}\{\hat{m}(x)\}}_{\text{sampling noise}}$$

have the same asymptotic order.

Prediction and inference are different tasks

A bandwidth that is optimal for mean squared prediction error does not automatically justify a confidence interval centered at $\hat{m}(x)$.

Leave-one-out errors reveal overfitting

In practice, ideally, we hope to select h minimize the IMSE:

$$IMSE_n(h) = \int E[(\hat{m}(x, h) - m(x))^2]w(x)dx$$

Ordinary residuals are misleading because $h \rightarrow 0$ implies

$$\hat{m}(X_i) \rightarrow Y_i, \quad \hat{e}_i = Y_i - \hat{m}(X_i) \rightarrow 0.$$

Instead, omit observation i when predicting its outcome:

$$\tilde{e}_i(h) = Y_i - \hat{m}_{-i}(X_i; h).$$

Select

$$\hat{h}_{CV} = \arg \min_{h \geq h_{\min}} CV(h) = \frac{1}{n} \sum_{i=1}^n \tilde{e}_i(h)^2 w(X_i).$$

Up to a constant that does not depend on h ,

$$\mathbb{E}[CV(h)] \approx IMSE_{n-1}(h).$$

Conditional variance is another unknown function

Conditional variance is $\sigma^2(x) = \text{Var}[Y|X = x] = \mathbb{E}[e^2|X = x]$.

Estimate residuals using leave-one-out predictions:

$$\tilde{e}_i = Y_i - \hat{m}_{-i}(X_i).$$

A kernel estimator of the conditional variance is

$$\hat{\sigma}^2(x) = \frac{\sum_i K\left(\frac{X_i - x}{h_\sigma}\right) \tilde{e}_i^2}{\sum_i K\left(\frac{X_i - x}{h_\sigma}\right)}.$$

- h_σ need not equal the bandwidth used to estimate the conditional mean.
- It can be selected by cross-validation applied to squared prediction errors.
- Smoothing squared in-sample residuals can understate variance when the mean fit is flexible.

The sampling variance to be estimated

For a degree- p local polynomial, define

$$Z_i(x) = (1, X_i - x, \dots, (X_i - x)^p)', \quad K_i = K\left(\frac{X_i - x}{h}\right).$$

Stack $Z_i(x)'$ into Z_x , set $W_x = \text{diag}(K_1, \dots, K_n)$, and write $A_x = (Z_x' W_x Z_x)^{-1}$ when the inverse exists. Then

$$\hat{\beta}(x) = A_x Z_x' W_x \mathbf{Y}, \quad \hat{m}(x) = \hat{\beta}_0(x).$$

Conditional on $\mathbf{X}$, this is linear in the independent errors. Hence

$$V_{\beta}(x) := \text{Var}(\hat{\beta}(x) \mid \mathbf{X}) = A_x \left[\sum_i K_i^2 Z_i(x) Z_i(x)' \sigma^2(X_i) \right] A_x.$$

- $\sigma^2(X_i)$ measures noise in Y_i ; $V_{\beta}(x)$ measures uncertainty in $\hat{\beta}(x)$.
- The first diagonal entry is $\text{Var}(\hat{m}(x) \mid \mathbf{X})$. NW uses $p = 0$; local linear regression uses $p = 1$.

Hansen, *Econometrics*, §19.16, p. 690.

Estimator 1: the White-type sandwich

Replace the unknown $\sigma^2(X_i)$ by a squared residual or prediction error:

$$\hat{e}_i = Y_i - \hat{m}(X_i), \quad \tilde{e}_i = Y_i - \hat{m}_{-i}(X_i).$$

Using leave-one-out prediction errors gives

$$\hat{V}_{\beta,1}(x) = A_x \left[\sum_i K_i^2 Z_i(x) Z_i(x)' \tilde{e}_i^2 \right] A_x.$$

For NW, $Z_i(x) = 1$, so this reduces to the scalar formula

$$\hat{V}_{m,1}(x) = \frac{\sum_i K_i^2 \tilde{e}_i^2}{(\sum_i K_i)^2}.$$

- Replacing $\tilde{e}_i^2$ with $\hat{e}_i^2$ gives the usual residual-based version.
- Hansen recommends prediction errors: each Y_i is predicted by a fit that excludes observation i .

Hansen, *Econometrics*, §19.16, equation (19.18), pp. 690–691.

Estimator 2: smoothed conditional variances

Smooth squared prediction errors to estimate $\sigma^2(t)$ with bandwidth h_σ . Evaluate this function at each X_i and substitute into the covariance formula:

$$\hat{V}_{\beta,2}(x) = A_x \left[\sum_i K_i^2 Z_i(x) Z_i(x)' \hat{\sigma}^2(X_i) \right] A_x.$$

Another version uses the target-point estimate $\hat{\sigma}^2(x)$ for all observations in the local fit:

$$\hat{V}_{\beta,2}^{\text{local}}(x) = \hat{\sigma}^2(x) A_x (Z'_x W_x^2 Z_x) A_x.$$

- Smoothing replaces individual squared errors by estimates of their conditional means.
- Evaluating at x approximates the noise variance as constant within the shrinking neighborhood.
- The additional bandwidth h_σ controls how much these variance estimates are smoothed.

Hansen, *Econometrics*, §19.16, p. 691; the conditional variance estimator is equation (19.17).

Estimator 3: the asymptotic plug-in formula

For NW and local linear regression at an interior point, substitute estimates into the leading variance formula:

$$\widehat{V}_{m,3}(x) = \frac{R_K \widehat{\sigma}^2(x)}{nh \widehat{f}_b(x)}, \quad R_K = \int K(u)^2 du.$$

Here $\widehat{\sigma}^2(x)$ is the smoothed conditional variance, and

$$\widehat{f}_b(x) = \frac{1}{nb} \sum_{i=1}^n K\left(\frac{X_i - x}{b}\right).$$

The mean bandwidth h , density bandwidth b , and variance bandwidth h_σ need not coincide. This approach uses the limiting variance expression in place of the full local design matrix.

For estimators 1–2, take the first diagonal entry of $\widehat{V}_{\beta,j}(x)$. For all three choices, the standard error is

$$\widehat{\text{se}}_j\{\widehat{m}(x)\} = \sqrt{\widehat{V}_{m,j}(x)}, \quad j = 1, 2, 3.$$

Hansen, *Econometrics*, §19.16, p. 691, including equation (19.19). Hansen's recommended choice is estimator 1 with leave-one-out prediction errors.

A conventional interval is pointwise

A nominal 95% interval at one pre-specified point is

$$\hat{m}(x) \pm 1.96 \hat{\text{se}}\{\hat{m}(x)\}.$$

This statement concerns repeated samples at that fixed value of x .

Two qualifications

1. The standard error captures sampling variance, not smoothing bias. It centers on $m(X) + h^2 B(x)$.
2. Pointwise 95% intervals drawn at many values of x do not jointly cover the entire curve with probability 95%.

These distinctions apply to kernel, series, and spline estimators.

Uniform bands answer a stronger question

A pointwise interval asks whether it covers $m(x)$ at one fixed x . A uniform band seeks

$$\Pr(m(x) \in [\hat{m}(x) \pm c_{1-\alpha} \widehat{\text{se}}\{\hat{m}(x)\}] \text{ for every } x \in \mathcal{X}_0) \approx 1 - \alpha.$$

- The simultaneous critical value $c_{1-\alpha}$ is larger than 1.96.
- It depends on the maximum fluctuation of the studentized process over the evaluation set.
- Simulation or bootstrap methods are typically used in applications.
- Bias control remains necessary.

Key idea

Use a uniform band when the claim concerns the shape of the entire function, such as monotonicity or a threshold.

Product kernels handle several predictors

For $X \in \mathbb{R}^d$, define

$$K_{\mathbf{h}}(X_i - x) = \prod_{j=1}^d K\left(\frac{X_{ij} - x_j}{h_j}\right), \quad |\mathbf{h}| = \prod_{j=1}^d h_j.$$

With local linear regression and suitable conditions,

$$\sqrt{n|\mathbf{h}|} \left[\hat{m}(x) - m(x) - \sum_{j=1}^d h_j^2 B_j(x) \right] \xrightarrow{d} N\left(0, \frac{R_K^d \sigma^2(x)}{f(x)}\right).$$

For local linear regression,

$$B_j(x) = \frac{1}{2} \frac{\partial^2 m(x)}{\partial x_j^2}.$$

The curse of dimensionality empties neighborhoods

With a common bandwidth h in d dimensions,

$$\text{AIMSE}(h) = O\left(h^4 + \frac{1}{nh^d}\right).$$

Therefore,

$$h_{\text{opt}} \asymp n^{-1/(4+d)}, \quad \text{AIMSE}_{\text{opt}} = O\left(n^{-4/(4+d)}\right).$$

Dimension d	1	2	4	10
AIMSE exponent $4/(4+d)$	0.80	0.67	0.50	0.29

Key idea

With many continuous predictors, useful structure is not optional: locality alone cannot overcome sparse data.

Series regression uses flexible basis features

Let

$$P_K(x) = \begin{pmatrix} p_1(x) & \cdots & p_K(x) \end{pmatrix}', \quad m_K(x) = P_K(x)' \beta_K.$$

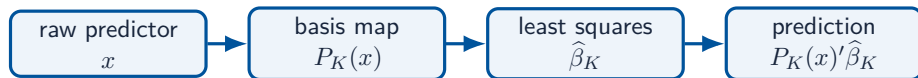
Define the population projection coefficient

$$\beta_K = \mathbb{E}[P_K(X)P_K(X)']^{-1} \mathbb{E}[P_K(X)Y].$$

Estimate it by least squares:

$$\hat{\beta}_K = \left(\sum_{i=1}^n P_K(X_i)P_K(X_i)' \right)^{-1} \left(\sum_{i=1}^n P_K(X_i)Y_i \right), \quad \hat{m}_K(x) = P_K(x)' \hat{\beta}_K.$$

The ML view: features, then linear algebra



- The fitted function is nonlinear in the raw predictor x .
- It remains linear in the coefficients β_K .
- The basis determines the representation; K determines the model's complexity.
- As n grows, allowing K to grow turns a finite regression into a nonparametric estimator.

Key idea

Series regression is a transparent example of representation learning through engineered features.

Polynomial regression is the simplest series estimator

A degree- p polynomial uses

$$P_K(x) = (1 \quad x \quad x^2 \quad \cdots \quad x^p)', \quad K = p + 1.$$

- Small p is parsimonious but potentially misspecified; large p reduces approximation error but raises estimation variance.
- Model complexity should grow with sample size rather than remain fixed.

Key idea

The same polynomial order can fit a large sample and overfit a smaller one.

High-order polynomials create two problems

Global influence

- Every coefficient affects the curve everywhere.
- Improving local fit can distort distant regions.
- Oscillation is often most severe near boundaries.

Ill-conditioning

- Powers of X can have very different scales.
- $X, X^2, \dots, X^p$ are highly correlated.
- $P'P$ may have a very large condition number.

Key idea

A polynomial basis can approximate smooth functions, but it is often a poor computational representation for flexible fitting.

Orthogonal polynomials improve conditioning

Replace raw powers with basis functions $q_0(x), \dots, q_p(x)$ that are orthogonal under a chosen inner product:

$$\mathbb{E}[q_j(X)q_k(X)] = 0 \quad \text{for } j \neq k.$$

- The orthogonal basis spans the same degree- p polynomial space.
- Fitted values are unchanged up to numerical precision.
- Coefficients are different because the coordinates have changed.
- Scaling X before constructing powers is a simpler partial remedy.

Key idea

Reparameterization can stabilize computation without changing the set of functions the model can represent.

Series error separates approximation from estimation

Let the projection approximation error be

$$r_K(x) = m(x) - m_K(x) = m(x) - P_K(x)' \beta_K.$$

Then

$$\hat{m}_K(x) - m(x) = \underbrace{P_K(x)'(\hat{\beta}_K - \beta_K)}_{\text{estimation error}} - \underbrace{r_K(x)}_{\text{approximation error}}.$$

- Increasing K enlarges the function space and reduces approximation error.
- Increasing K also requires estimating more coefficients and raises sampling variation.
- This is the series analogue of reducing the kernel bandwidth.

Smoothness controls approximation error

The expected squared projection error

$$\delta_K^2 = \mathbb{E}[r_K^2].$$

If $m^{(\alpha)}$ is uniformly continuous on a compact support and the polynomial or spline order is at least α , then

$$\delta_K = o(K^{-\alpha}).$$

In particular, when m'' is uniformly continuous,

$$\delta_K = O(K^{-2})$$

is available for linear splines, with the sharper little- o rate for polynomials and sufficiently high-order splines.

Key idea

Smoothness determines how quickly additional basis functions reduce approximation bias.

Consistency

- We know for any fixed K , $\hat{\beta}_K \rightarrow \beta_K$
- And thus $\hat{m}_K(x) = X_K(x)' \hat{\beta}_K \rightarrow X_K(x)' \beta_K$ as $n \rightarrow \infty$
- Furthermore, we know $X_K(x)' \beta_K \rightarrow m(x)$ as $k \rightarrow \infty$
- It therefore seems reasonable that $\hat{m}_K(x) \rightarrow m(X)$ as both $n \rightarrow \infty$ and $K \rightarrow \infty$.

Consistency in integrated squared error

Let F be the distribution of X . The integrated squared error is

$$\text{ISE}(K) = \int [\hat{m}_K(x) - m(x)]^2 dF(x).$$

Theorem 20.6: consistency (Section 20.11)

Under Assumption 20.1 and $\delta_K = o(1)$, as $n, K \rightarrow \infty$,

$$\boxed{\text{ISE}(K) = o_p(1).}$$

- Continuous m on a compact interval and $K \rightarrow \infty$ imply $\delta_K \rightarrow 0$ for polynomials.
- Fixed K generally estimates a projection m_K with nonzero approximation error.
- This is $L_2(F)$ consistency: squared error averaged over X vanishes in probability.

Hansen, *Econometrics*, §20.11, Theorem 20.6, p. 720; the usual i.i.d. regression setting with $\mathbb{E}[Y^2] < \infty$.

Theorem 20.7: the series convergence rate

Under Assumption 20.1 and bounded conditional variance $\sigma^2(x) \leq \bar{\sigma}^2 < \infty$, as $n, K \rightarrow \infty$,

$$\text{ISE}(K) = O_p\left(\delta_K^2 + \frac{K}{n}\right).$$

For polynomials with m'' uniformly continuous on a compact interval,

$$\text{ISE}(K) = O_p\left(\underbrace{K^{-4}}_{\text{squared approximation bias}} + \underbrace{\frac{K}{n}}_{\text{estimation variance}}\right).$$

- $\delta_K^2 = \mathbb{E}[(m(X) - m_K(X))^2]$ measures how well the basis approximates m .
- K/n is the variance cost of estimating K coefficients from n observations.

Hansen, *Econometrics*, §20.12, Theorem 20.7, p. 721. These are stochastic order bounds.

Polynomial degree trades bias against variance

Small K : a restrictive polynomial

Few coefficients give a stable fit, but the polynomial may miss curvature.
Approximation bias can dominate the error.

Large K : a flexible polynomial

More terms reduce approximation bias, but estimating more coefficients increases the variance cost K/n .

For the twice-smooth bound in Theorem 20.7, balance the two orders:

$$K^{-4} \asymp \frac{K}{n} \quad \implies \quad K \asymp n^{1/5}, \quad \text{ISE}(K) = O_p(n^{-4/5}).$$

This gives the growth rate that balances the bound. The appropriate constant depends on the regression function and the noise level.

Key idea

Increase degree $p = K - 1$ slowly with n ; select it by prediction validation.

Hansen, *Econometrics*, §20.12, p. 721.

Bandwidth and basis dimension control complexity

	Kernel / local regression	Series regression
Complexity	smaller h	larger K
Approximation bias	$O(h^2)$	$O(K^{-2})$
Variance	$O((nh)^{-1})$	$O(K/n)$
Optimal tuning	$h \asymp n^{-1/5}$	$K \asymp n^{1/5}$
Optimal integrated MSE	$O(n^{-4/5})$	$O(n^{-4/5})$
Computational character	fit locally at prediction	fit coefficients once

Key idea

Think of K^{-1} as a series bandwidth, or h^{-1} as the effective number of local parameters.

Series estimators center on the finite-basis projection

Let

$$\Omega_K = \mathbb{E}[P_K(X)P_K(X)'e^2], \quad V_K(x) = P_K(x)'Q_K^{-1}\Omega_K Q_K^{-1}P_K(x).$$

Under increasing-dimension regularity conditions,

$$\frac{\sqrt{n} [\hat{m}_K(x) - m(x) + r_K(x)]}{V_K(x)^{1/2}} \xrightarrow{d} N(0, 1).$$

- The variance has the same sandwich form as ordinary least squares.
- The approximation error $r_K(x)$ is the asymptotic bias.
- The effective pointwise rate is slower than $\sqrt{n}$ because $V_K(x)$ generally grows with K .

Series undersmoothing uses extra basis functions

If the uniform approximation error satisfies

$$n(\delta_K^*)^2 \rightarrow 0,$$

then

$$\frac{\sqrt{n} [\hat{m}_K(x) - m(x)]}{V_K(x)^{1/2}} \xrightarrow{d} N(0, 1).$$

When m'' is uniformly continuous, $K \asymp n^{1/4}$ is sufficient for polynomials and quadratic splines—faster growth than the ISE-optimal $K \asymp n^{1/5}$.

- More terms reduce approximation bias.
- More terms increase estimation variance and can worsen conditioning.
- As with kernels, the formal condition does not itself provide a practical tuning rule.

Splines replace global polynomials with local pieces

A spline is a piecewise polynomial joined at points called *knots*.

Degree	Piecewise form	Usual smoothness restriction
0	constants	jumps allowed
1	lines	function is continuous
2	quadratics	function and first derivative are continuous
3	cubics	function, first, and second derivatives are continuous

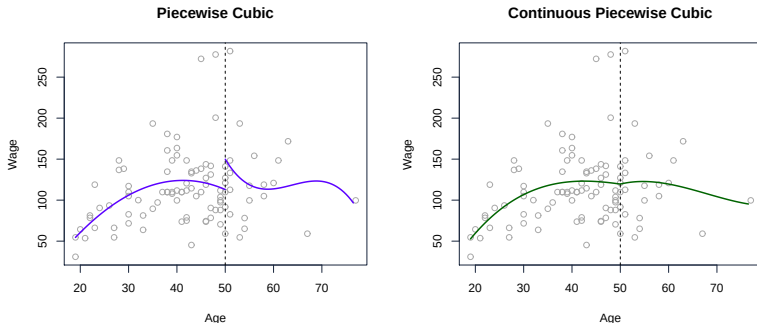
- Polynomial degree controls smoothness within each segment.
- Knots determine where the shape may adapt.
- Low-degree pieces avoid the oscillation of a high-degree global polynomial.

Key idea

Splines achieve flexibility through more local pieces, not through ever-higher polynomial degree.

Joining cubic pieces: continuity at the knot

The same wage data are fit on either side of a knot at age 50.

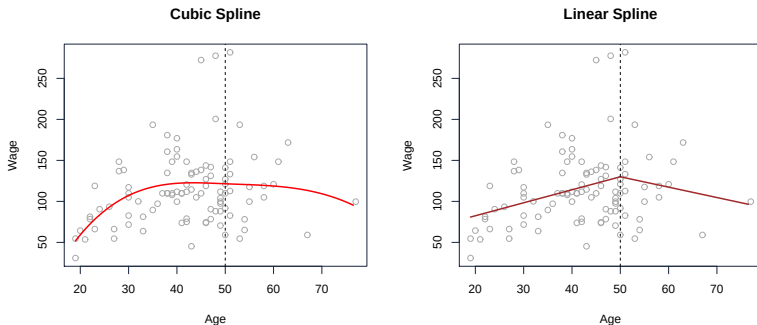


- **Left:** two unrestricted cubics use 8 coefficients and can jump at the knot.
- **Right:** matching their values imposes one restriction (7 coefficients); their slopes can still differ.

James, Witten, Hastie, and Tibshirani, *An Introduction to Statistical Learning*, Figure 7.3 (top row), p. 276.

Smoothness constraints define the spline

Further restrictions determine how smoothly the pieces join at age 50.



- **Cubic spline:** match the function and its first two derivatives: $8 - 3 = 5$ coefficients.
- **Linear spline:** match the two lines' values; a kink remains: $4 - 1 = 3$ coefficients.

James, Witten, Hastie, and Tibshirani, *An Introduction to Statistical Learning*, Figure 7.3 (bottom row), p. 276.

A linear spline makes the slope change interpretable

With one knot τ ,

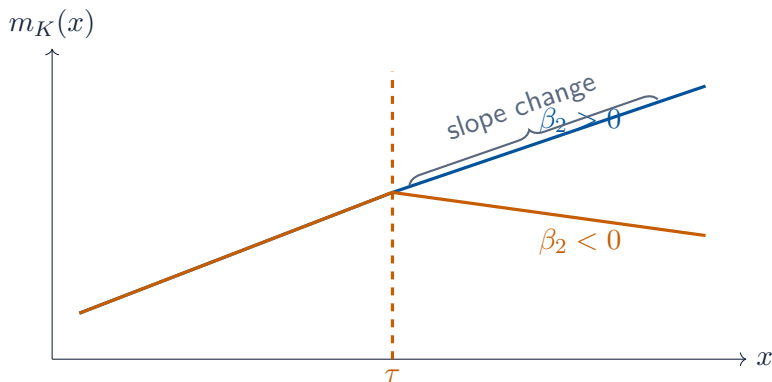
$$m_K(x) = \beta_0 + \beta_1 x + \beta_2(x - \tau)_+, \quad (x - \tau)_+ = \max\{x - \tau, 0\}.$$

Therefore,

$$m'_K(x) = \begin{cases} \beta_1, & x < \tau, \\ \beta_1 + \beta_2, & x > \tau. \end{cases}$$

- β_2 is the change in slope at the knot.
- Continuity is automatic because $(x - \tau)_+ = 0$ at $x = \tau$.
- Linear splines are useful when theory permits a sharp change in marginal effects.

A knot adds local flexibility



Key idea

The truncated basis is zero before the knot, so its coefficient changes only the portion of the function to the right.

The truncated-power basis builds a spline

A degree- p spline with interior knots $\tau_1 < \dots < \tau_N$ can be written

$$m_K(x) = \sum_{j=0}^p \beta_j x^j + \sum_{k=1}^N \gamma_k (x - \tau_k)_+^p, \quad K = N + p + 1.$$

- The global polynomial terms establish the baseline shape.
- Each truncated term permits a local change beginning at its knot.
- A cubic spline is twice continuously differentiable and has $N + 4$ coefficients.
- Once the basis is constructed, estimation is ordinary least squares.

Key idea

Smoothness constraints are encoded in the basis rather than imposed through a nonlinear estimation algorithm.

Knot placement allocates the model's flexibility

Equal spacing

Reasonable when X is approximately uniform and the same spatial resolution is desired throughout the support.

Quantile spacing

Places more knots where observations are dense and fewer where data are sparse; common software default.

Substantive placement

Uses known thresholds or regions of rapid curvature, but risks data-driven specification search.

- The number of knots or degrees of freedom is the main complexity parameter.
- Choose complexity with validation or cross-validation.

Key idea

Knots say where the model may bend; cross-validation helps decide how many opportunities to bend it needs.

Natural cubic splines stabilize the boundaries

Ordinary cubic splines can have high variance near the extreme predictor values. A natural cubic spline imposes

$$m_K''(x) = 0 \quad \text{outside the boundary knots,}$$

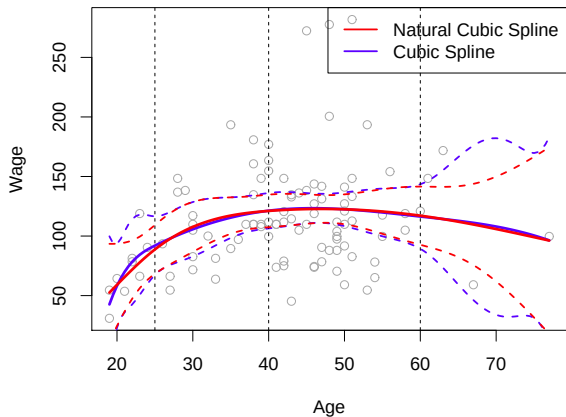
so the fitted function is linear in both tails.

- The restrictions trade a small amount of flexibility for much more stable tail behavior.
- Linear tails are safer for modest extrapolation, but extrapolation still rests on an assumption.

Key idea

Boundary restrictions are a form of regularization informed by where nonparametric estimators are weakest.

Natural cubic splines: the boundary comparison



- The fitted means are similar in the interior.
- The ordinary cubic spline's dashed confidence curves spread out more at the edges.
- Natural boundary restrictions stabilize the fit where observations are sparse.

James, Witten, Hastie, and Tibshirani, *An Introduction to Statistical Learning*, Figure 7.4, p. 278: ordinary and natural cubic splines with three interior knots, fit to a subset of the Wage data.

A regression-spline workflow has four explicit choices

1. **Select degree:** usually linear for interpretable kinks or cubic for smooth curvature.
2. **Select candidate complexity:** knot counts or equivalent degrees of freedom.
3. **Place knots:** often at sample quantiles within the training fold.
4. **Select complexity:** minimize validation or cross-validation error, then refit on the training data.

Avoid data leakage

If knot locations depend on sample quantiles, recompute those quantiles inside each training fold rather than using the full dataset before cross-validation.

Key idea

The fitted coefficients are easy; the central modeling decisions concern representation and complexity.

Smoothing splines penalize roughness

Estimate the entire function by solving

$$\hat{f}_\lambda = \arg \min_f \left\{ \sum_{i=1}^n [Y_i - f(X_i)]^2 + \lambda \int [f''(t)]^2 dt \right\}.$$

Data-fit term

Rewards a function that predicts the observed outcomes closely.

Roughness penalty

Penalizes curvature; a straight line has zero penalty.

Key idea

This is the same loss-plus-penalty principle as ridge and LASSO, applied to the shape of a function.

The solution is finite-dimensional

A central representer result states that the minimizer is a natural cubic spline with knots at the distinct observed predictor values. Using a spline basis matrix B and curvature-penalty matrix Ω ,

$$\hat{\beta}_\lambda = (B'B + \lambda\Omega)^{-1}B'Y,$$

and

$$\hat{f}_\lambda = B\hat{\beta}_\lambda = S_\lambda Y, \quad S_\lambda = B(B'B + \lambda\Omega)^{-1}B'.$$

- The rich knot set supplies potential flexibility.
- The penalty, rather than discrete knot selection, determines how much flexibility is used.

Effective degrees of freedom measure smoothness

$\lambda = 0$

The curve interpolates the observations; effective degrees of freedom approach the number of distinct X_i values.

Intermediate λ

Curved components are continuously shrunk, producing a smooth nonlinear fit.

$\lambda \rightarrow \infty$

Curvature is eliminated; the solution approaches the least-squares line with two degrees of freedom.

Key idea

Effective degrees of freedom describe flexibility continuously, even when the underlying basis contains many coefficients.

Smoothing splines have fast exact leave-one-out CV

The leave-one-out criterion is

$$\text{CV}(\lambda) = \frac{1}{n} \sum_{i=1}^n \left[\frac{Y_i - \hat{f}_\lambda(X_i)}{1 - (S_\lambda)_{ii}} \right]^2.$$

Generalized cross-validation replaces individual leverages by their average:

$$\text{GCV}(\lambda) = \frac{n^{-1} \|\mathbf{Y} - \hat{\mathbf{f}}_\lambda\|^2}{[1 - \text{df}(\lambda)/n]^2}.$$

- Both criteria can be evaluated without fitting n separate models.
- GCV is rotation-invariant and often computationally convenient.

Two spline strategies regularize differently

	Regression spline	Smoothing spline
Candidate knot set	modest, chosen explicitly	all distinct X_i values
Fit	unpenalized least squares	penalized least squares
Complexity control	knots or degrees of freedom	λ or effective df
Tuning	CV over discrete models	CV/GCV over a continuous path
Boundary behavior	ordinary or natural basis	natural cubic spline solution
Link to prior methods	feature engineering	generalized ridge regression

Key idea

Regression splines restrict the size of the function space; smoothing splines keep a rich space and shrink its rough components.

The central lessons

1. Nonparametric estimation replaces a fixed functional form with smoothness and complexity choices.
2. Kernel and local-polynomial estimators learn through nearby observations; bandwidth controls locality.
3. Series and spline estimators learn through basis features; K or λ controls complexity.
4. Prediction-optimal tuning does not automatically remove smoothing or approximation bias from inference.
5. The curse of dimensionality motivates structured models such as additive models and modern regularized ML.

Key idea

Every flexible estimator needs three arguments: why its representation is appropriate, how its complexity is selected, and what uncertainty statement is justified.

References I